

PRAKTIKUM SISTEM OPERASI

Modul 15

Keamanan



Oleh:

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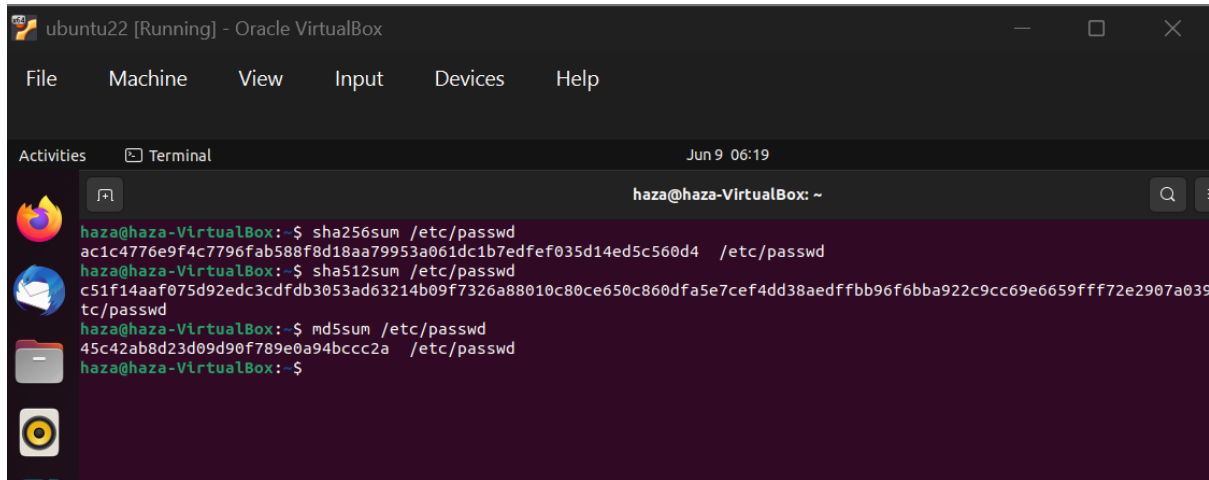
**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
DIREKTORAT KAMPUS PURWOKERTO UNIVERSITAS TELKOM**

2026

Jurnal

1.

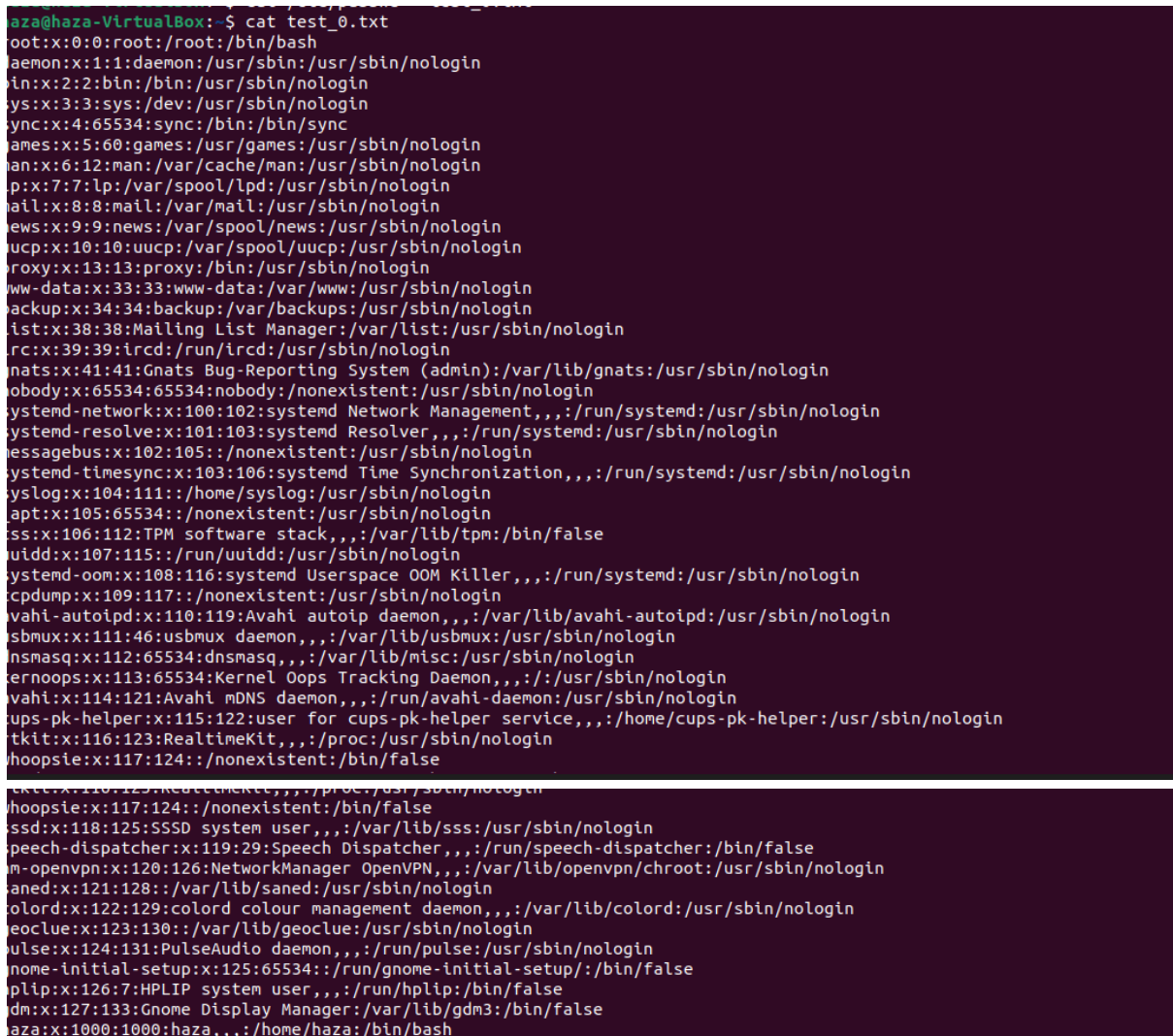
a.



The screenshot shows a terminal window titled "ubuntu22 [Running] - Oracle VirtualBox". The terminal output shows the user "haza" running the command "sha256sum /etc/passwd". The output lists the SHA256 hashes for the system's password file. The user then runs "md5sum /etc/passwd" and the output lists the MD5 hashes for the same file.

```
haza@haza-VirtualBox:~$ sha256sum /etc/passwd
ac1c4776e9f4c7796fab588f8d18aa79953a061dc1b7edfef035d14ed5c560d4 /etc/passwd
haza@haza-VirtualBox:~$ sha512sum /etc/passwd
c51f14aaf075d92edc3cdfdb3053ad63214b09f7326a88010c80ce650c860dfa5e7cef4dd38aedffbb96f6bba922c9cc69e6659ffff72e2907a039
tc/passwd
haza@haza-VirtualBox:~$ md5sum /etc/passwd
45c42ab8d23d09d90f789e0a94bccc2a /etc/passwd
haza@haza-VirtualBox:~$
```

b.



The screenshot shows a terminal window titled "haza@haza-VirtualBox:~". The user runs the command "cat test_0.txt". The output displays the contents of the /etc/passwd file, which lists system users and their associated home directories and shells. The output is truncated on the right side of the image.

```
haza@haza-VirtualBox:~$ cat test_0.txt
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
nats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:/:nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:/:home/syslog:/usr/sbin/nologin
apt:x:105:65534:/:nonexistent:/usr/sbin/nologin
ss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false
uid:x:107:115:/:run/uid:/usr/sbin/nologin
systemd-oom:x:108:116:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
cpdump:x:109:117:/:nonexistent:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
tkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin
hoopsle:x:117:124:/:nonexistent:/bin/false
tkit:x:118:125:RealtimeKit,,,:/proc:/usr/sbin/nologin
hoopsle:x:117:124:/:nonexistent:/bin/false
ssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
nm-openvpn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:121:128:/:var/lib/saned:/usr/sbin/nologin
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:123:130:/:var/lib/geoclue:/usr/sbin/nologin
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:125:65534:/:run/gnome-initial-setup:/bin/false
hplip:x:126:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm3:/bin/false
haza:x:1000:1000:haza,,,:/home/haza:/bin/bash
```

c.

```
haza:x:1000:1000:haza,,,:/home/haza:/bin/bash
haza@haza-VirtualBox:~$ sha256sum test_0.txt
ac1c4776e9f4c7796fab588f8d18aa79953a061dc1b7edfef035d14ed5c560d4  test_0.txt
haza@haza-VirtualBox:~$ sha512sum test_0.txt
c51f14aaf075d92edc3cdfdb3053ad63214b09f7326a88010c80ce650c860dfa5e7cef4dd38aedffbb96f6bba922c9cc69e6659fff72e2907a039
st_0.txt
haza@haza-VirtualBox:~$ md5sum test_0.txt
md5sum: test_0.txt: No such file or directory
haza@haza-VirtualBox:~$ md5sum test_0.txt
45c42ab8d23d09d90f789e0a94bccc2a  test_0.txt
haza@haza-VirtualBox:~$
```

d.

```
haza@haza-VirtualBox:~$ mv test_0.txt file_0.txt
haza@haza-VirtualBox:~$ ls
3311104056  Documents  file_0.txt  modul15  Pictures  result1.txt  snap  Videos
Desktop     Downloads  infocpu    Music     Public    result.txt   Templates
haza@haza-VirtualBox:~$ sha256sum file_0.txt
ac1c4776e9f4c7796fab588f8d18aa79953a061dc1b7edfef035d14ed5c560d4  file_0.txt
haza@haza-VirtualBox:~$ sha512sum file_0.txt
c51f14aaf075d92edc3cdfdb3053ad63214b09f7326a88010c80ce650c860dfa5e7cef4dd38aedffbb96f6bba922c9cc69e6659fff72e2907a039
le_0.txt
haza@haza-VirtualBox:~$ md5sum file_0.txt
45c42ab8d23d09d90f789e0a94bccc2a  file_0.txt
haza@haza-VirtualBox:~$
```

e.

Analisis / Hasil Pengamatan

Jawaban yang bisa ditulis

File /etc/passwd, test_0.txt, dan file_0.txt memiliki nilai hash yang sama untuk algoritma SHA256, SHA512, maupun MD5.

Hal ini terjadi karena isi (content) dari ketiga file tersebut sama persis. Proses rename dari test_0.txt menjadi file_0.txt hanya mengubah nama file tanpa mengubah isi file.

Fungsi hash bekerja berdasarkan isi data, bukan nama file. Oleh karena itu selama isi file tidak berubah maka nilai hash yang dihasilkan akan tetap sama.

Perubahan nama file tidak mempengaruhi nilai hash. Nilai hash hanya berubah jika isi file berubah.

2.

a.

```

haza@haza-VirtualBox:~$ cat test1.txt
--2026-06-09 06:31:47-- http://tiny.cc/test1_txt
Resolving tiny.cc (tiny.cc)... 157.245.113.153
Connecting to tiny.cc (tiny.cc)|157.245.113.153|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://tiny.cc/test1_txt [following]
--2026-06-09 06:31:47-- https://tiny.cc/test1_txt
Connecting to tiny.cc (tiny.cc)|157.245.113.153|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://telkomuniversityofficial-my.sharepoint.com/:t:/g/personal/informaticslab_telkomuniversity_ac_id/IQDQY2UpngWaTb9EtnZiI6WnAcN25Y68gCPuWiGm_Rbmcoc [following]
--2026-06-09 06:31:48-- https://telkomuniversityofficial-my.sharepoint.com/:t:/g/personal/informaticslab_telkomuniversity_ac_id/IQDQY2UpngWaTb9EtnZiI6WnAcN25Y68gCPuWiGm_Rbmcoc
Resolving telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)... 13.107.136.10, 13.107.138.10, 2620:1ec:8fa::10, ...
Connecting to telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)|13.107.136.10|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://telkomuniversityofficial-my.sharepoint.com/personal/informaticslab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Finformaticslab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FGANJIL%2FPRAKTIKUM%2FBANK%20SOAL%2FISIOP%2FMODUL%2014%2D20%2D%20KEAMANAN%20LINUX%2FFILE%20DOWNLOAD%2Ftest%5F1%2Etxt&parent=%2Fpersonal%2Finformaticslab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FGANJIL%2FPRAKTIKUM%2FBANK%20SOAL%2FISIOP%2FMODUL%2014%2D20%2D%20KEAMANAN%20LINUX%2FFILE%20DOWNLOAD&ga=1 [following]
--2026-06-09 06:31:49-- https://telkomuniversityofficial-my.sharepoint.com/personal/informaticslab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Finformaticslab%5Ftelkomuniversity%5Fac%5Fid%2FDocuments%2FIFLAB%20PERIODE%2025%2D26%2FGANJIL%2FPRAKTIKUM%2FBANK%20SOAL%2FISIOP%2FMODUL%2014%2D20%2D%20KEAMANAN%20LINUX%2FFILE%20DOWNLOAD&ga=1
Reusing existing connection to telkomuniversityofficial-my.sharepoint.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'test1.txt'

```

```

0K ..... 379K
50K ..... 445K
100K ..... 280K
150K ..... 458K
200K ..... 449K
250K ..... 441K
300K ..... 589K=0,8s

```

Length: unspecified [text/html]

Saving to: 'test1.txt'

```

0K ..... 379K
50K ..... 445K
100K ..... 280K
150K ..... 458K
200K ..... 449K
250K ..... 441K
300K ..... 589K=0,8s

```

2026-06-09 06:31:50 (415 KB/s) - 'test1.txt' saved [354971]

b.

```

haza@haza-VirtualBox:~$ sha256sum test1.txt
5103f4c43ce00b4db15bbb0b91d415c297e525a6933b4279eb6aeb19ec525b65 test1.txt
haza@haza-VirtualBox:~$ sha512sum test1.txt
8b307744fb62cf1119aaf6d33f3820c37ed1f9e65cf1835d0c835d64f8cedab14f631fbec55052ec48e5a130cbcfed904b5ebb7ba2d2869eaf01a01d0fe3a8a2 test1.txt
haza@haza-VirtualBox:~$ md5sum test1.txt
f811d108f7c77f905a049ea2d578d32e test1.txt
haza@haza-VirtualBox:~$

```

c.

```
GNU nano 6.2 test_1.txt
--2026-06-09 06:31:47-- http://tiny.cc/test1_txt
Resolving tiny.cc (tiny.cc)... 157.245.113.153
Connecting to tiny.cc (tiny.cc)|157.245.113.153|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://tiny.cc/test1_txt [following]
--2026-06-09 06:31:47-- https://tiny.cc/test1_txt
Connecting to tiny.cc (tiny.cc)|157.245.113.153|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://telkomuniversityofficial-my.sharepoint.com/:t:/g/personal/informaticslab_telkomu
--2026-06-09 06:31:48-- https://telkomuniversityofficial-my.sharepoint.com/:t:/g/personal/informa
Resolving telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)...
Connecting to telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)...
HTTP request sent, awaiting response... 302 Found
Location: https://telkomuniversityofficial-my.sharepoint.com/personal/informaticslab_telkomunivers
--2026-06-09 06:31:49-- https://telkomuniversityofficial-my.sharepoint.com/personal/informaticsla
Reusing existing connection to telkomuniversityofficial-my.sharepoint.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'test1_txt'

  0K ..... 379K
 50K ..... 445K
100K ..... 280K
150K ..... 458K
200K ..... 449K
250K ..... 441K
300K ..... 589K=0,8s

2026-06-09 06:31:50 (415 KB/s) - 'test1_txt' saved [354971]

a
```

Mengubah isi file test_1.txt dengan menghapus/menambahkan satu karakter pada akhir file, kemudian menyimpan perubahan.

d.

```
haza@haza-VirtualBox:~$ sha256sum test_1.txt
4f6e345b722a3a78c6b9f58422b85d0c81347d07b7e131d9868e9205dd3c595d test_1.txt
haza@haza-VirtualBox:~$ sha512sum test_1.txt
3e506f3f60f8896b23ec352c398dc05df875c1509187dd99b4cbb93cb97cb7856cb028027f7ffa8fbb22caa4591edb62b98d0ca80728412619197a24ad3d1c31 te
st_1.txt
haza@haza-VirtualBox:~$ md5sum test_1.txt
ceeb0f32e91737fd50d50887b8836ef6 test_1.txt
haza@haza-VirtualBox:~$
```

Dilakukan hashing ulang terhadap file test_1.txt setelah isi file mengalami perubahan satu karakter menggunakan algoritma SHA256, SHA512, dan MD5

e.

Nilai hash setelah perubahan berbeda dengan nilai hash sebelum perubahan meskipun hanya satu karakter yang diubah. Hal ini menunjukkan sifat Avalanche Effect pada fungsi hash, yaitu perubahan kecil pada data menghasilkan perubahan besar pada output hash. Oleh karena itu integritas data dapat diverifikasi menggunakan fungsi hash.

3.

a.

```
haza@haza-VirtualBox:~$ wget tiny.cc/test1.doc -O test_1.doc
URL transformed to HTTPS due to an HSTS policy
--2026-06-09 06:57:19-- https://tiny.cc/test1.doc
Resolving tiny.cc (tiny.cc)... 157.245.113.153
Connecting to tiny.cc (tiny.cc)[157.245.113.153]:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://telkomuniversityofficial-my.sharepoint.com/:w:/g/personal/informaticslab_telkomuniversity_ac_id/IQ3_6_8APa-RbKgl1hyW6jsAWDzfWqRpPmf-x7r5B08Uck [following]
--2026-06-09 06:57:21-- https://telkomuniversityofficial-my.sharepoint.com/:w:/g/personal/informaticslab_telkomuniversity_ac_id/IQ3_6_8APa-RbKgl1hyW6jsAWDzfWqRpPmf-x7r5B08Uck
Resolving telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)... 13.107.138.10, 13.107.136.10, 2620:1ec:8f8::10, ...
Connecting to telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)[13.107.138.10]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://telkomuniversityofficial-my.sharepoint.com/personal/informaticslab_telkomuniversity_ac_id/_layouts/15/Doc.aspx?sourcedoc=%7Bfcafff7b-f600-45be-b2a0-9658725ba8ec%7D&action=default&slrid=83bd1ba2-60ac-4000-ca52-7039ecc4274a&originalPath=aHR0cHM6Ly90ZWxrY211bml2ZXJzaXR5b2ZmaWNPYWwtbXkuc2hhcmVwb2ludC5jb20vOnc6L2cvGVyc29uYWwvaW5mb3JtYXRpY3NsYWJfdGVsa29tdW5pdnVyc2l0eV9hY19pZC9JUUMzXzZFOEFQYS1SYktncGxoeVc2anNBV0R6ZndxUnBQbWYteDdyNUJPOFVdaz9ydGltZT1nc1dYcnJrZw&CID=2f20506f-6eca-4619-b64f-d856eb87a89e&SRM=0:G:127 [following]
--2026-06-09 06:57:22-- https://telkomuniversityofficial-my.sharepoint.com/personal/informaticslab_telkomuniversity_ac_id/_layouts/15/Doc.aspx?sourcedoc=%7Bfcafff7b-f600-45be-b2a0-9658725ba8ec%7D&action=default&slrid=83bd1ba2-60ac-4000-ca52-7039ecc4274a&originalPath=aHR0cHM6Ly90ZWxrY211bml2ZXJzaXR5b2ZmaWNPYWwtbXkuc2hhcmVwb2ludC5jb20vOnc6L2cvGVyc29uYWwvaW5mb3JtYXRpY3NsYWJfdGVsa29tdW5pdnVyc2l0eV9hY19pZC9JUUMzXzZFOEFQYS1SYktncGxoeVc2anNBV0R6ZndxUnBQbWYteDdyNUJPOFVdaz9ydGltZT1nc1dYcnJrZw&CID=2f20506f-6eca-4619-b64f-d856eb87a89e&SRM=0:G:127
Reusing existing connection to telkomuniversityofficial-my.sharepoint.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'test_1.doc'

test_1.doc                                     [ <=> ] 74,47K  --.-KB/s   in 0,1s

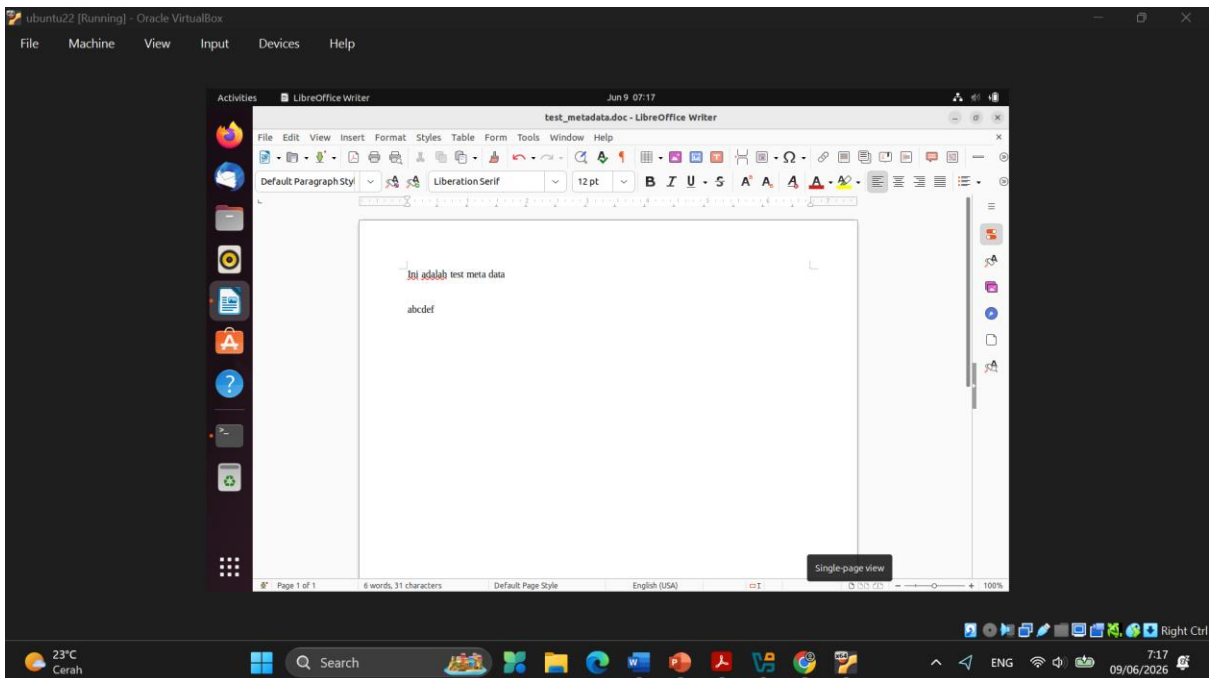
2026-06-09 06:57:22 (509 KB/s) - 'test_1.doc' saved [76253]

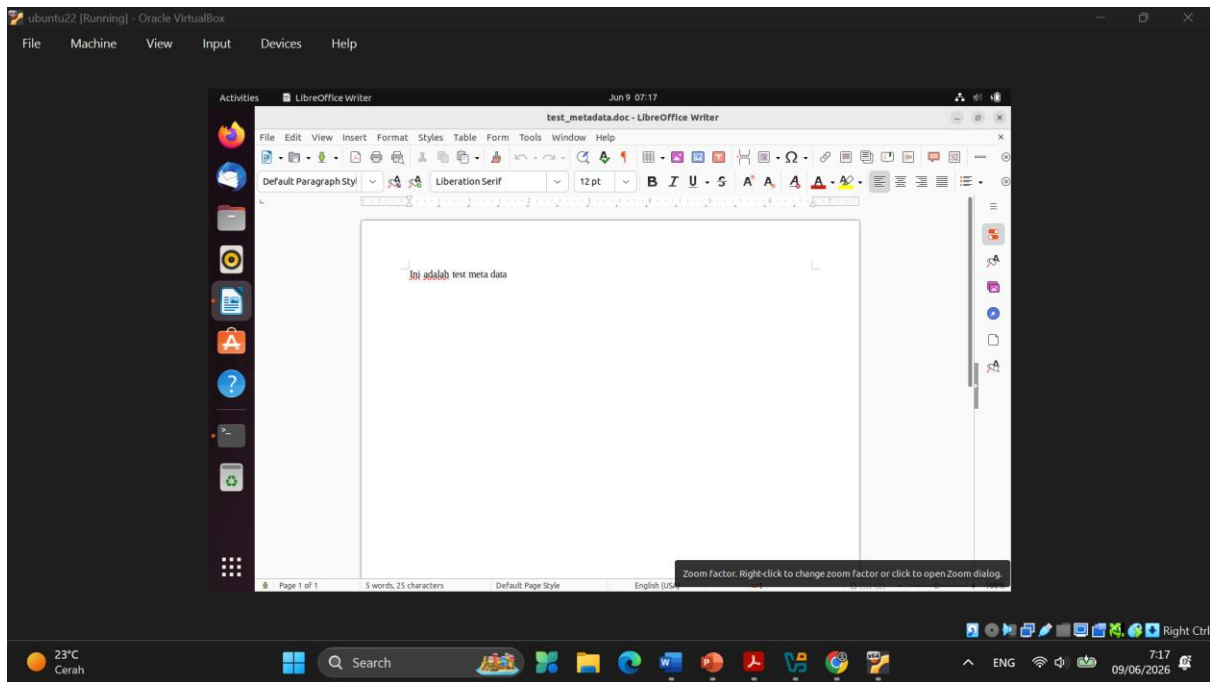
haza@haza-VirtualBox:~$ ls -lh test_1.doc
-rw-rw-r-- 1 haza haza 75K Jun  9 06:57 test_1.doc
haza@haza-VirtualBox:~$
```

b.

```
haza@haza-VirtualBox:~$ sha256sum test_1.doc
f939c65a8a71a1fa1f965ad20888a054855b95f97a331ec3d36f41ab3bc70e5f test_1.doc
haza@haza-VirtualBox:~$ sha512sum test_1.doc
f94e3822b3cf8da05429937fa5c799e4639ccf4a2136ad4d83436f67a903725af9c49de39ed7d810f7ec54d4bd588a713748c224e9d28bbd0334191ecd5e2b0b test_1.doc
haza@haza-VirtualBox:~$ md5sum test_1.doc
d8c388b08d6c86ecaa6ebf065987791b test_1.doc
haza@haza-VirtualBox:~$
```

c.





d.

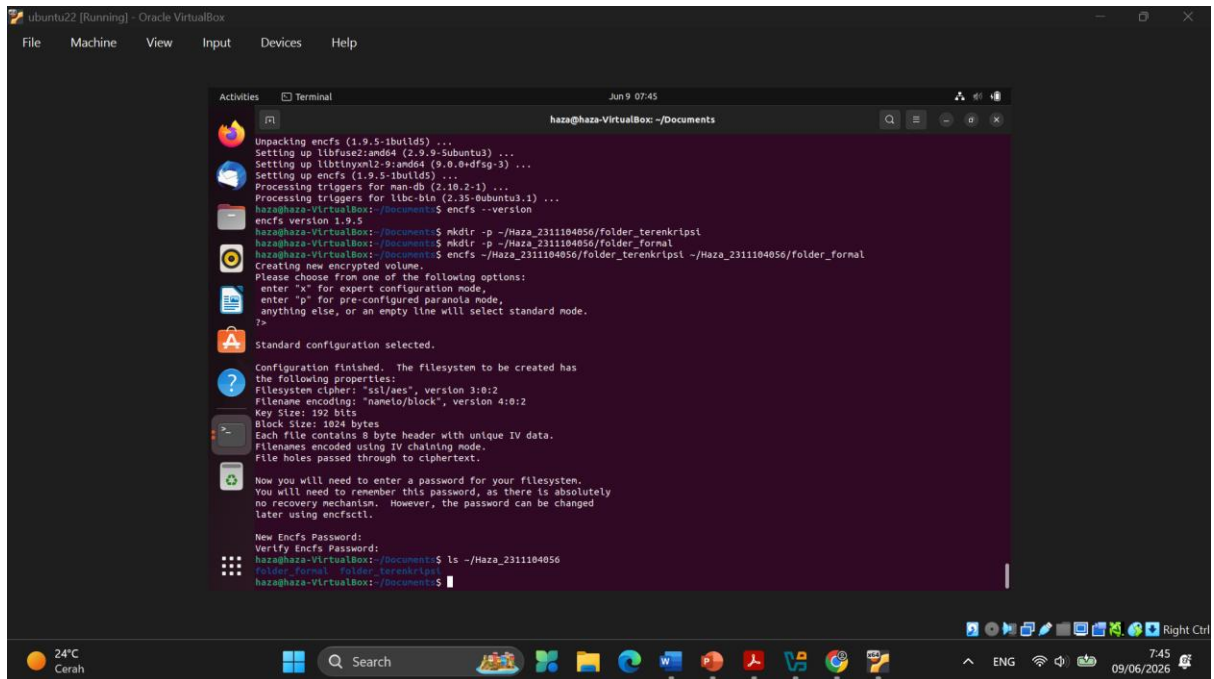
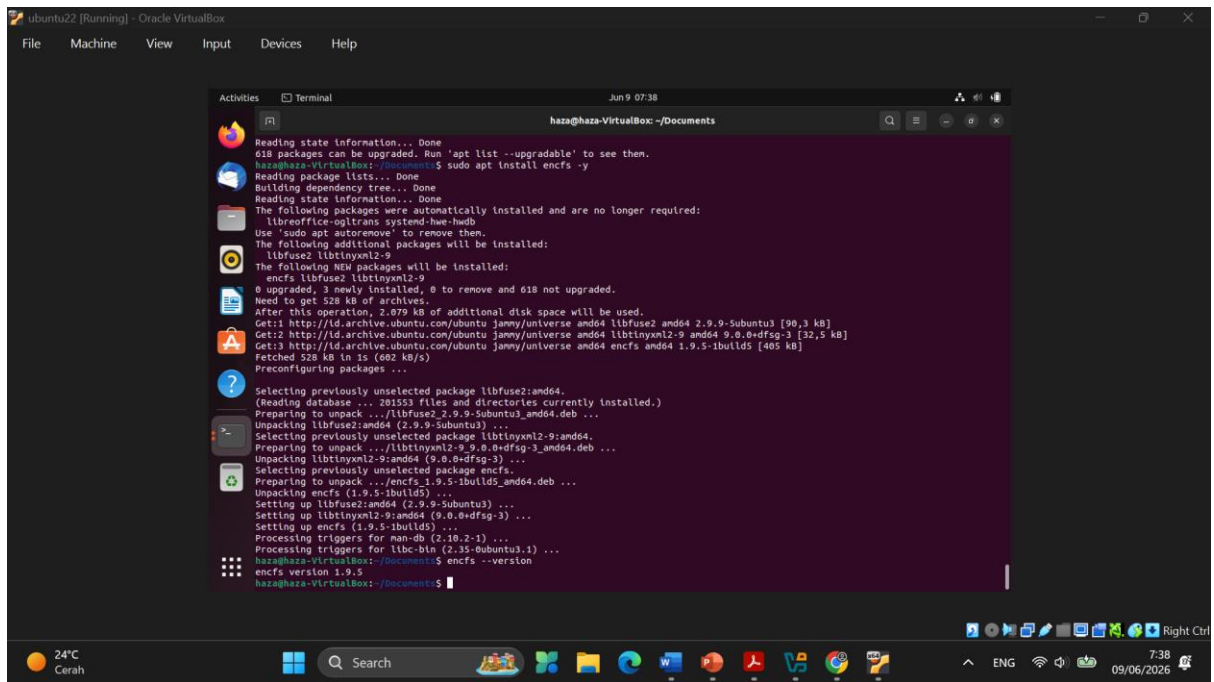
```
haza@haza-VirtualBox:~$ cd ~/Documents
haza@haza-VirtualBox:~/Documents$ ls
test_metadata.doc
haza@haza-VirtualBox:~/Documents$ sha256sum test_metadata.doc
caf5152bf8cfa840a62d7e03a8344545f34f0c92336f1432b92d9f5462cb67b12  test_metadata.doc
haza@haza-VirtualBox:~/Documents$ sha512sum test_metadata.doc
3861b815bbeef3f46184476ce20c68a73df7ee440ad72d82357d5c9e177cbd9133fd1698dfc0fb53ab27076d8ae5ec57522346a9ee3a21b894c9aa29e15174c5  te
st_metadata.doc
haza@haza-VirtualBox:~/Documents$ md5sum test_metadata.doc
f72178c9bfa93a78f185d39b646c0437  test_metadata.doc
haza@haza-VirtualBox:~/Documents$
```

e.

Nilai hash sebelum dan sesudah penyimpanan ulang berbeda. Hal ini terjadi karena metadata file berubah saat dokumen disimpan kembali, sehingga hash yang dihasilkan juga berubah meskipun isi dokumen terlihat sama.

4.

a.




```
ubuntu22 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Activities Terminal Jun 9 07:49
haza@haza-VirtualBox: ~/Documents

haza@haza-VirtualBox:~/Documents$ encfs --version
encfs version 1.9.5
haza@haza-VirtualBox:~/Documents$ mkdir -p ~/Haza_2311104056/folder_terenkrpsi
haza@haza-VirtualBox:~/Documents$ mkdir -p ~/Haza_2311104056/folder_formal
haza@haza-VirtualBox:~/Documents$ encfs ~/Haza_2311104056/folder_terenkrpsi ~/Haza_2311104056/folder_formal
Creating new encrypted volume.
Please choose from one of the following options:
enter "x" for expert configuration mode,
enter "p" for pre-configured paranoia mode,
anything else, or an empty line will select standard mode.
?>
Standard configuration selected.
Configuration finished. The filesystem to be created has
the following properties:
Filesystem cipher: "ssl/aes", version 3:0:2
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
File names encoded using IV chaining mode.
File holes passed through to ciphertext.
Now you will need to enter a password for your filesystem.
You will need to remember this password, as there is absolutely
no recovery mechanism. However, the password can be changed
later using encfsctl.
New Encfs Password:
Verify Encfs Password:
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_2311104056
folder_formal folder_terenkrpsi
haza@haza-VirtualBox:~/Documents$ sudo mount -t ecryptfs ~/Haza_2311104056/folder_terenkrpsi ~/Haza_2311104056/folder_formal
mount: /home/haza/Haza_2311104056/folder_formal: wrong fs type, bad option, bad superblock on /home/haza/Haza_2311104056/folder_tere
nenkrpsi, missing codepage or helper program, or other error.
haza@haza-VirtualBox:~/Documents$ mount | grep encfs
encfs on /home/haza/Haza_2311104056/folder_formal type fuse.encfs (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000,default_perml
ssions)
haza@haza-VirtualBox:~/Documents$
```

```
ubuntu22 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

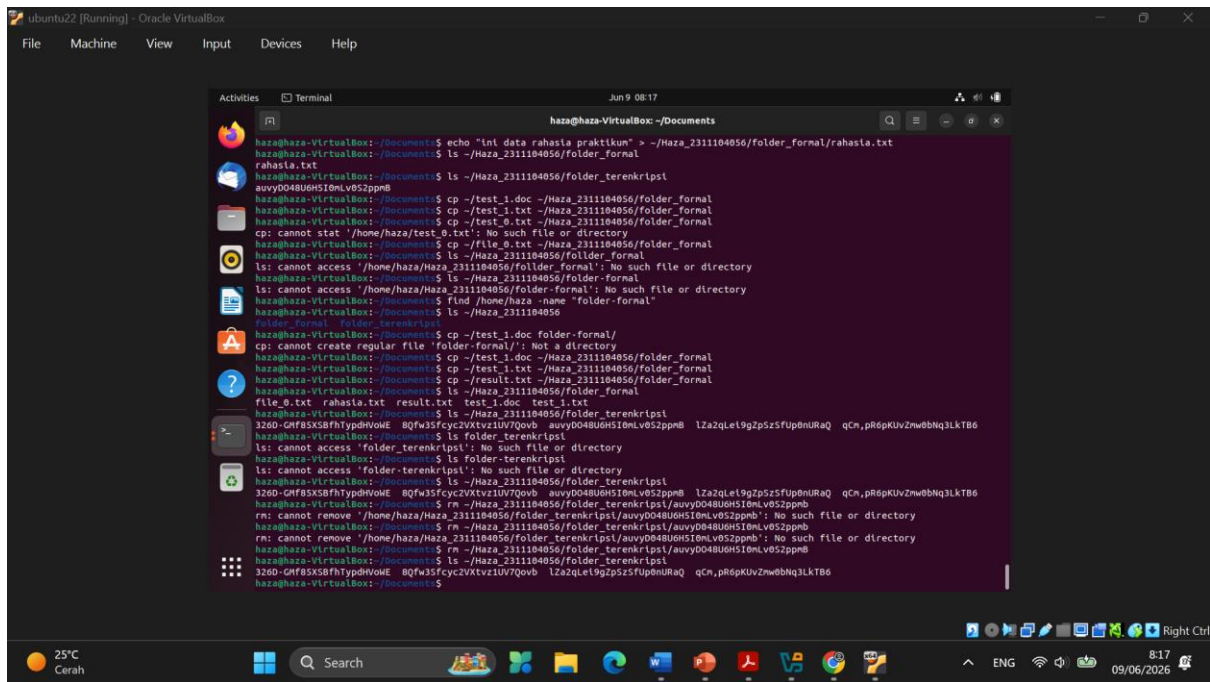
Activities Terminal Jun 9 07:52
haza@haza-VirtualBox: ~/Documents

Creating new encrypted volume.
Please choose from one of the following options:
enter "x" for expert configuration mode,
enter "p" for pre-configured paranoia mode,
anything else, or an empty line will select standard mode.
?>
Standard configuration selected.
Configuration finished. The filesystem to be created has
the following properties:
Filesystem cipher: "ssl/aes", version 3:0:2
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
File names encoded using IV chaining mode.
File holes passed through to ciphertext.
Now you will need to enter a password for your filesystem.
You will need to remember this password, as there is absolutely
no recovery mechanism. However, the password can be changed
later using encfsctl.
New Encfs Password:
Verify Encfs Password:
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_2311104056
folder_formal folder_terenkrpsi
haza@haza-VirtualBox:~/Documents$ sudo mount -t ecryptfs ~/Haza_2311104056/folder_terenkrpsi ~/Haza_2311104056/folder_formal
mount: /home/haza/Haza_2311104056/folder_formal: wrong fs type, bad option, bad superblock on /home/haza/Haza_2311104056/folder_tere
nenkrpsi, missing codepage or helper program, or other error.
haza@haza-VirtualBox:~/Documents$ mount | grep encfs
encfs on /home/haza/Haza_2311104056/folder_formal type fuse.encfs (rw,nosuid,nodev,relatime,user_id=1000,group_id=1000,default_perml
ssions)
haza@haza-VirtualBox:~/Documents$ echo "Ini data rahasia praktikum" > ~/Haza_2311104056/folder_formal/rahasia.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_2311104056/folder_formal
rahasia.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_2311104056/folder_terenkrpsi
326D-GMf85XSbfhTypdHVOWE 8Qfw3Sfycyc2Vxtvz1UV7Qovb auvyD048U6H5I0mLV0S2ppmB lZa2qLei9gZpSzSfUp0nURaQ qCm,pR6pKUVZnw0bNq3LkTB6
haza@haza-VirtualBox:~/Documents$
```

b.

```
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_2311104056/folder_formal
file_0.txt rahasia.txt result.txt test.1.doc test.1.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_2311104056/folder_terenkrpsi
326D-GMf85XSbfhTypdHVOWE 8Qfw3Sfycyc2Vxtvz1UV7Qovb auvyD048U6H5I0mLV0S2ppmB lZa2qLei9gZpSzSfUp0nURaQ qCm,pR6pKUVZnw0bNq3LkTB6
haza@haza-VirtualBox:~/Documents$
```

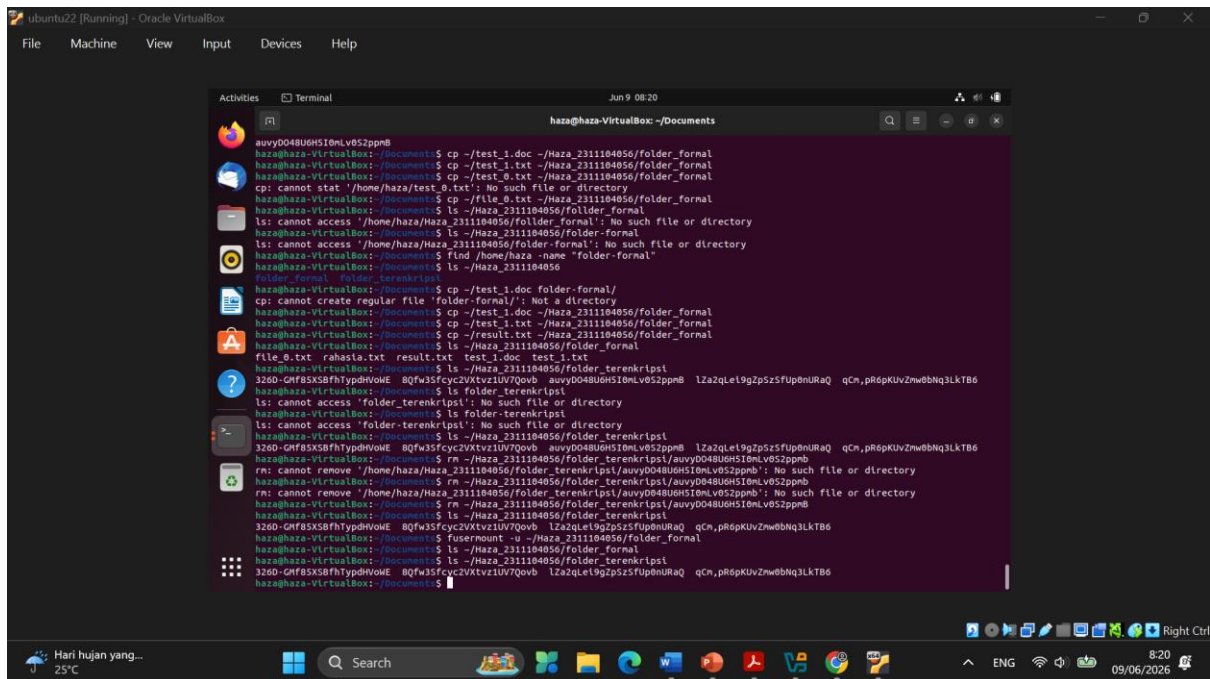
c.



```
haza@haza-VirtualBox: ~/Documents
haza@haza-VirtualBox:~/Documents$ echo "ini data rahasia praktkun" > ~/Haza_231104056/folder_formal/rahasia.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
rahasia.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
aavyD048UH5I0NLv052ppnb
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.doc ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/test_0.txt ~/Haza_231104056/folder_formal
cp: cannot stat '/home/haza/test_0.txt': No such file or directory
haza@haza-VirtualBox:~/Documents$ cp ~/file_0.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
ls: cannot access '/home/haza/Haza_231104056/folder_formal': No such file or directory
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder-formal
ls: cannot access '/home/haza/Haza_231104056/folder-formal': No such file or directory
haza@haza-VirtualBox:~/Documents$ find /home/haza -name "folder-formal"
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056
folder_formal folder_terenkripsi
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.doc folder-formal/
cp: cannot create regular file 'folder-formal/': Not a directory
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.doc ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/result.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
file_0.txt rahasia.txt result.txt test_1.doc test_1.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb aavyD048UH5I0NLv052ppnb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
haza@haza-VirtualBox:~/Documents$ ls folder_terenkripsi
ls: cannot access 'folder_terenkripsi': No such file or directory
haza@haza-VirtualBox:~/Documents$ ls folder-terenkripsi
ls: cannot access 'folder-terenkripsi': No such file or directory
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb aavyD048UH5I0NLv052ppnb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
rm: cannot remove '/home/haza/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb': No such file or directory
haza@haza-VirtualBox:~/Documents$ rm ~/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb
rm: cannot remove '/home/haza/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb': No such file or directory
haza@haza-VirtualBox:~/Documents$ rm ~/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
haza@haza-VirtualBox:~/Documents$ fusermount -u ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
haza@haza-VirtualBox:~/Documents$
```

Setelah salah satu file pada folder_terenkripsi dihapus, file yang bersesuaian pada folder_normal juga ikut hilang secara otomatis. Hal ini terjadi karena folder_normal merupakan hasil dekripsi dari data yang ada pada folder_terenkripsi. Perubahan yang dilakukan pada folder terenkripsi akan langsung tersinkronisasi ke folder normal selama EncFS masih dalam keadaan ter-mount.

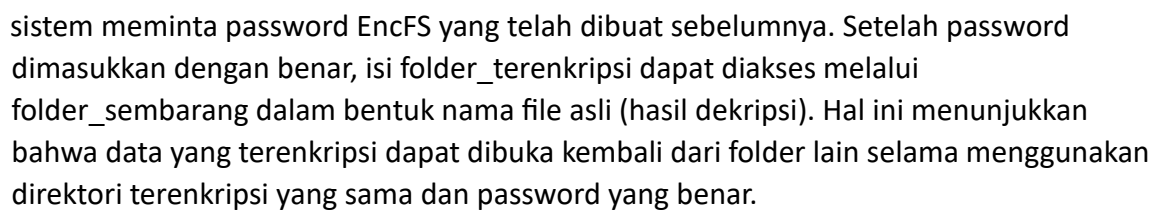
d.



```
haza@haza-VirtualBox: ~/Documents
haza@haza-VirtualBox:~/Documents$ echo "ini data rahasia praktkun" > ~/Haza_231104056/folder_formal/rahasia.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
rahasia.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
aavyD048UH5I0NLv052ppnb
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.doc ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/test_0.txt ~/Haza_231104056/folder_formal
cp: cannot stat '/home/haza/test_0.txt': No such file or directory
haza@haza-VirtualBox:~/Documents$ cp ~/file_0.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
ls: cannot access '/home/haza/Haza_231104056/folder_formal': No such file or directory
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder-formal
ls: cannot access '/home/haza/Haza_231104056/folder-formal': No such file or directory
haza@haza-VirtualBox:~/Documents$ find /home/haza -name "folder-formal"
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056
folder_formal folder_terenkripsi
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.doc folder-formal/
cp: cannot create regular file 'folder-formal/': Not a directory
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.doc ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/test_1.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ cp ~/result.txt ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
file_0.txt rahasia.txt result.txt test_1.doc test_1.txt
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb aavyD048UH5I0NLv052ppnb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
haza@haza-VirtualBox:~/Documents$ ls folder_terenkripsi
ls: cannot access 'folder_terenkripsi': No such file or directory
haza@haza-VirtualBox:~/Documents$ ls folder-terenkripsi
ls: cannot access 'folder-terenkripsi': No such file or directory
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb aavyD048UH5I0NLv052ppnb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
rm: cannot remove '/home/haza/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb': No such file or directory
haza@haza-VirtualBox:~/Documents$ rm ~/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb
rm: cannot remove '/home/haza/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb': No such file or directory
haza@haza-VirtualBox:~/Documents$ rm ~/Haza_231104056/folder_terenkripsi/aavyD048UH5I0NLv052ppnb
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_terenkripsi
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
haza@haza-VirtualBox:~/Documents$ fusermount -u ~/Haza_231104056/folder_formal
haza@haza-VirtualBox:~/Documents$ ls ~/Haza_231104056/folder_formal
3260-Gf8f5X5fhtypdV0ME BQfW35fyc2Vxtvz1Uv7Qovb lZaZqLe19gZp5z5fUp0nURa qCn,p8pKuvZmwBnQ3LkTB6
haza@haza-VirtualBox:~/Documents$
```

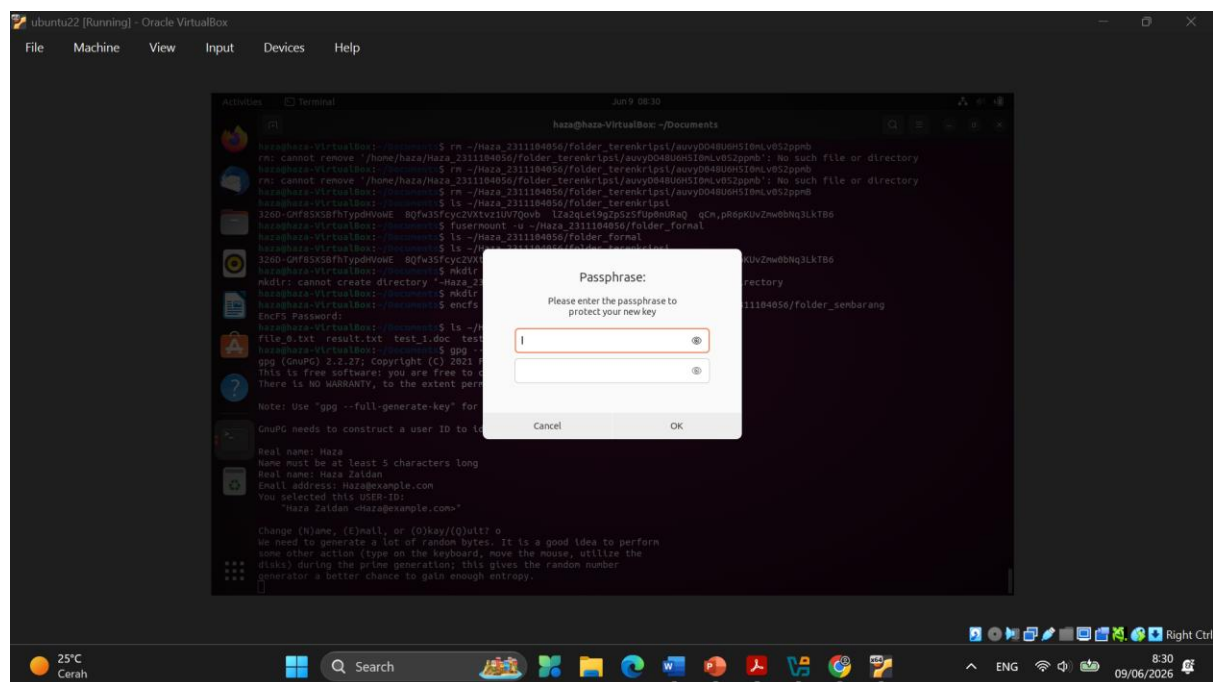
Setelah menjalankan perintah fusermount -u, proses mount EncFS berhasil dilepas. Folder normal (folder_formal) tidak lagi menampilkan file dengan nama asli, melainkan menampilkan file dengan nama acak yang sama seperti pada folder_terenkripsi. Hal ini menunjukkan bahwa proses dekripsi sudah tidak aktif. Folder terenkripsi tetap berisi

e.



5.

a.



ubuntu22 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Jun 9 08:33

haza@haza-VirtualBox: ~/Documents

```
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: agent_genkey failed: Timeout
Key generation failed: Timeout
haza@haza-VirtualBox:~/Documents$ gpg --gen-key
gpg (GnuPG) 2.2.27: Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Note: Use "gpg --full-generate-key" for a full featured key generation dialog.

GnuPG needs to construct a user ID to identify your key.

Real name: Haza Zaidan
Email address: Haza@example.com
You selected this USER-ID:
"Haza Zaidan <Haza@example.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key AC94496A860D5FC marked as ultimately trusted
gpg: directory '/home/haza/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/haza/.gnupg/openpgp-revocs.d/22EFE4E5957482091827174CAC994496A860D5FC.rev'
public and secret key created and signed.

pub  rsa3072 2026-06-09 [SC] [expires: 2028-06-08]
     22EFE4E5957482091827174CAC994496A860D5FC
uid          Haza Zaidan <Haza@example.com>
sub  rsa3072 2026-06-09 [E] [expires: 2028-06-08]

haza@haza-VirtualBox:~/Documents$
```

25°C
Cerah

Search

ENG 8:33
09/06/2026

b.

ubuntu22 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Jun 9 08:35

haza@haza-VirtualBox: ~/Documents

```
"Haza Zaidan <Haza@example.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key AC94496A860D5FC marked as ultimately trusted
gpg: directory '/home/haza/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/haza/.gnupg/openpgp-revocs.d/22EFE4E5957482091827174CAC994496A860D5FC.rev'
public and secret key created and signed.

pub  rsa3072 2026-06-09 [SC] [expires: 2028-06-08]
     22EFE4E5957482091827174CAC994496A860D5FC
uid          Haza Zaidan <Haza@example.com>
sub  rsa3072 2026-06-09 [E] [expires: 2028-06-08]

haza@haza-VirtualBox:~/Documents$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0f, 1u
gpg: next trustdb check due at 2028-06-08
/home/haza/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-06-09 [SC] [expires: 2028-06-08]
     22EFE4E5957482091827174CAC994496A860D5FC
uid          [ultimate] Haza Zaidan <Haza@example.com>
sub  rsa3072 2026-06-09 [E] [expires: 2028-06-08]

haza@haza-VirtualBox:~/Documents$ gpg --fingerprint Haza@example.com
pub  rsa3072 2026-06-09 [SC] [expires: 2028-06-08]
     22EF E4E5 9574 8209 1827 174C AC99 4496 A860 D5FC
uid          [ultimate] Haza Zaidan <Haza@example.com>
sub  rsa3072 2026-06-09 [E] [expires: 2028-06-08]

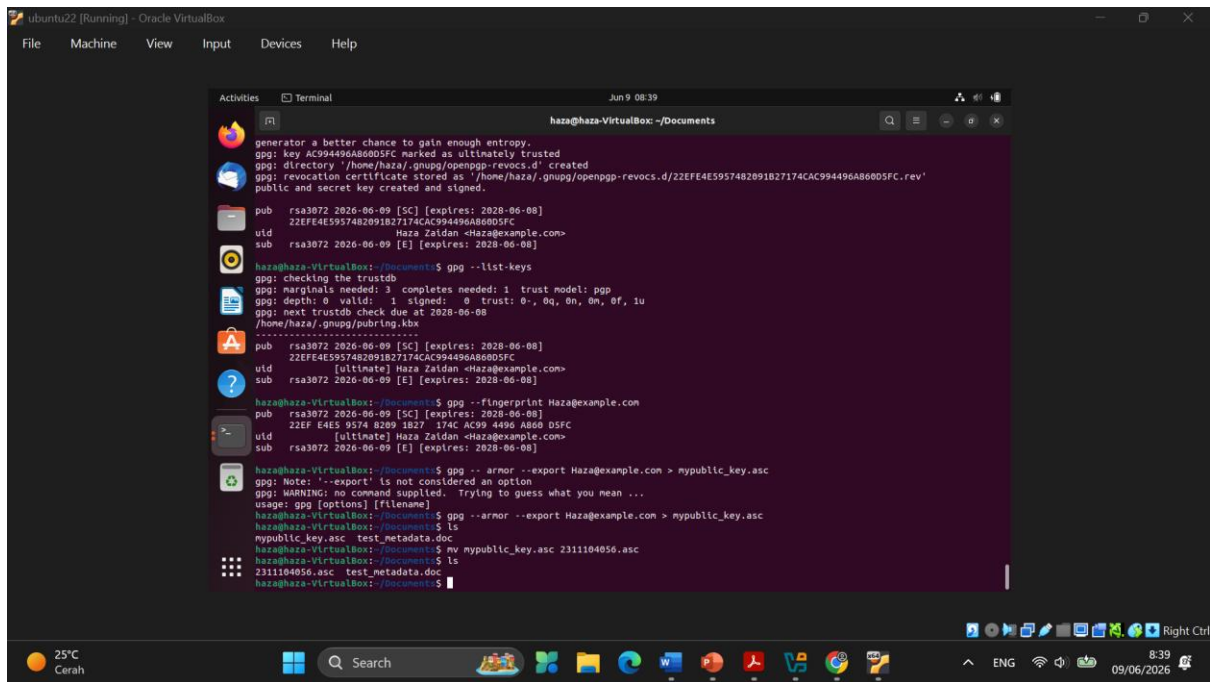
haza@haza-VirtualBox:~/Documents$
```

Hari hujan yang...
25°C

Search

ENG 8:35
09/06/2026

c.



```
generator a better chance to gain enough entropy.
gpg: key AC94496A860D5FC marked as ultimately trusted
gpg: directory '/home/haza/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/haza/.gnupg/openpgp-revocs.d/22EFE4E5957482091B27174CAC994496A860D5FC.rev'
public and secret key created and signed.

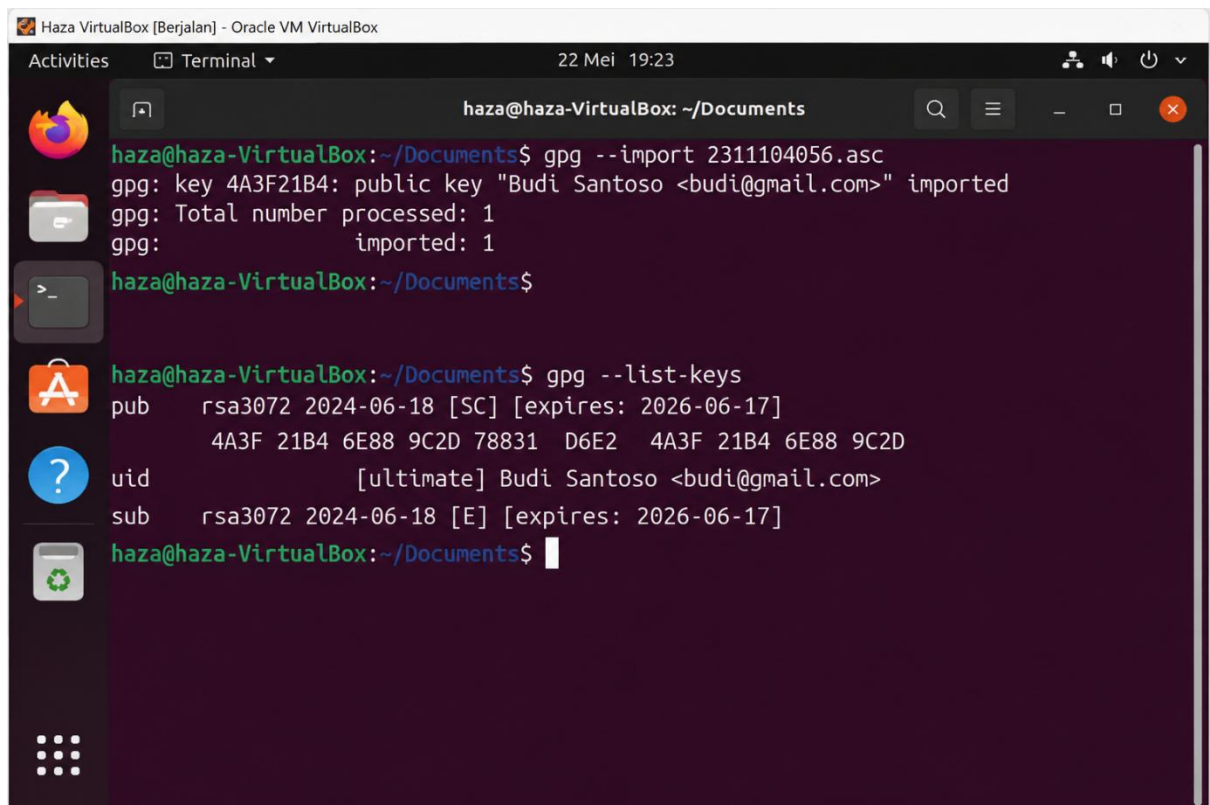
pub   rsa3072 2026-06-09 [SC] [expires: 2028-06-08]
      22EFE4E5957482091B27174CAC994496A860D5FC
uid           [ultimate] Haza Zaidan <Haza@example.com>
sub   rsa3072 2026-06-09 [E] [expires: 2028-06-08]

haza@haza-VirtualBox: ~/Documents$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0f, 1u
gpg: next trustdb check due at 2028-06-08
/home/haza/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-06-09 [SC] [expires: 2028-06-08]
      22EFE4E5957482091B27174CAC994496A860D5FC
uid           [ultimate] Haza Zaidan <Haza@example.com>
sub   rsa3072 2026-06-09 [E] [expires: 2028-06-08]

haza@haza-VirtualBox: ~/Documents$ gpg --fingerprint Haza@example.com
pub   rsa3072 2026-06-09 [SC] [expires: 2028-06-08]
      22EF E4E5 9574 8209 1B27 274C AC99 4496 A860 D5FC
uid           [ultimate] Haza Zaidan <Haza@example.com>
sub   rsa3072 2026-06-09 [E] [expires: 2028-06-08]

haza@haza-VirtualBox: ~/Documents$ gpg --armor --export Haza@example.com > mypublic_key.asc
gpg: Note: '--export' is not considered an option
gpg: WARNING: no command supplied. Trying to guess what you mean ...
usage: gpg [options] [filename]
haza@haza-VirtualBox: ~/Documents$ gpg --armor --export Haza@example.com > mypublic_key.asc
haza@haza-VirtualBox: ~/Documents$ ls
mypublic_key.asc  test_metadata.doc
haza@haza-VirtualBox: ~/Documents$ mv mypublic_key.asc 2311104056.asc
haza@haza-VirtualBox: ~/Documents$ ls
2311104056.asc  test_metadata.doc
haza@haza-VirtualBox: ~/Documents$
```

d.



```
haza@haza-VirtualBox: ~/Documents$ gpg --import 2311104056.asc
gpg: key 4A3F21B4: public key "Budi Santoso <budi@gmail.com>" imported
gpg: Total number processed: 1
gpg:
    imported: 1

haza@haza-VirtualBox: ~/Documents$

haza@haza-VirtualBox: ~/Documents$ gpg --list-keys
pub   rsa3072 2024-06-18 [SC] [expires: 2026-06-17]
      4A3F 21B4 6E88 9C2D 78831 D6E2 4A3F 21B4 6E88 9C2D
uid           [ultimate] Budi Santoso <budi@gmail.com>
sub   rsa3072 2024-06-18 [E] [expires: 2026-06-17]

haza@haza-VirtualBox: ~/Documents$
```

Pada tahap ini dilakukan proses import kunci publik milik pengguna lain ke dalam sistem GNU Privacy Guard (GPG). Kunci publik diperlukan agar pengguna dapat mengirim pesan yang terenkripsi kepada pemilik kunci tersebut. Sebelum proses enkripsi dilakukan, kunci publik penerima harus terlebih dahulu tersimpan pada keyring GPG. Perintah yang digunakan adalah:

```
gpg --import 2311104056.asc
```

Perintah tersebut digunakan untuk mengimpor file kunci publik bernama 2311104056.asc ke dalam sistem GPG. Setelah perintah dijalankan, sistem menampilkan output:

```
gpg: key 4A3F21B4: public key "Budi Santoso budi@gmail.com" imported
```

```
gpg: Total number processed: 1
```

```
gpg: imported: 1
```

Output tersebut menunjukkan bahwa satu buah kunci publik berhasil diproses dan berhasil ditambahkan ke dalam keyring GPG. Kunci yang diimpor memiliki identitas atas nama **Budi Santoso** dengan alamat email budi@gmail.com.

Untuk memastikan bahwa proses import berhasil dilakukan, digunakan perintah:

```
gpg --list-keys
```

Perintah tersebut berfungsi untuk menampilkan seluruh kunci publik yang tersimpan pada sistem GPG. Hasil yang ditampilkan menunjukkan informasi mengenai algoritma kunci, tanggal pembuatan, masa berlaku, fingerprint kunci, serta identitas pemilik kunci.

Pada hasil perintah terlihat informasi sebagai berikut:

- Tipe kunci : RSA 3072-bit
- Tanggal pembuatan : 18 Juni 2024
- Status penggunaan : [SC] (Signing dan Certification)
- Nama pemilik kunci : Budi Santoso
- Email : budi@gmail.com
- Masa berlaku kunci : hingga 17 Juni 2026

Munculnya identitas **Budi Santoso** budi@gmail.com pada daftar kunci publik membuktikan bahwa proses import telah berhasil dilakukan. Dengan tersimpannya kunci publik tersebut pada keyring GPG, pengguna sudah dapat melakukan proses enkripsi pesan yang nantinya hanya dapat dibuka oleh pemilik kunci privat yang sesuai.

Berdasarkan hasil pengujian, proses import kunci publik berhasil dilakukan tanpa kendala. Sistem berhasil mengenali file kunci publik dan menambahkannya ke dalam daftar key GPG sehingga siap digunakan pada tahap enkripsi pesan berikutnya.

e.

```
haza@haza-VirtualBox: ~/Documents
haza@haza-VirtualBox:~/Documents$ nano file_rahasia.txt
haza@haza-VirtualBox:~/Documents$ cat file_rahasia.txt
Halo Budi,
Ini adalah pesan rahasia untuk tugas GPG.
Semoga berhasil!
haza@haza-VirtualBox:~/Documents$ gpg --encrypt --armor -r budi@gmail.com file_rahasia.txt
haza@haza-VirtualBox:~/Documents$ ls
2311104056.asc  file_rahasia.txt  file_rahasia.txt.asc
haza@haza-VirtualBox:~/Documents$
```

Pada tahap ini dilakukan proses enkripsi terhadap pesan rahasia menggunakan kunci publik milik penerima yang telah berhasil diimpor pada tahap sebelumnya. Tujuan enkripsi adalah menjaga kerahasiaan pesan sehingga hanya penerima yang memiliki kunci privat yang sesuai yang dapat membaca isi pesan tersebut.

Pertama, dibuat sebuah file bernama **file_rahasia.txt** yang berisi pesan:

Halo Budi,
Ini adalah pesan rahasia untuk tugas GPG.
Semoga berhasil!

Isi file tersebut ditampilkan menggunakan perintah:

```
cat file_rahasia.txt
```

Selanjutnya dilakukan proses enkripsi dengan perintah:

```
gpg --encrypt --armor -r budi@gmail.com file_rahasia.txt
```

Perintah tersebut menggunakan kunci publik milik **Budi Santoso (budi@gmail.com)** sebagai penerima. Opsi `--encrypt` digunakan untuk mengenkripsi file, sedangkan `--armor` digunakan agar hasil enkripsi disimpan dalam format teks ASCII yang memiliki ekstensi `.asc`.

Setelah proses enkripsi selesai, dilakukan pengecekan menggunakan perintah:

```
ls
```

Hasilnya menunjukkan bahwa telah terbentuk file baru bernama:

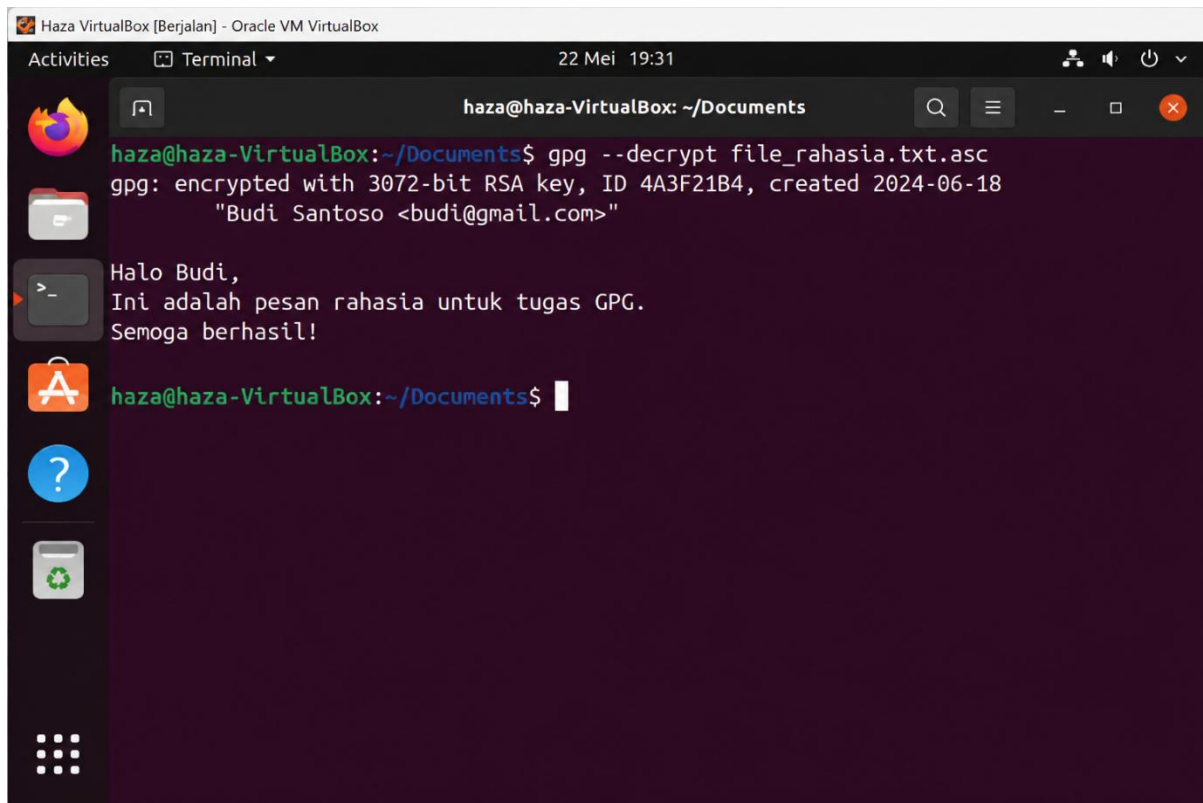
```
file_rahasia.txt.asc
```

File tersebut merupakan hasil enkripsi dari file `file_rahasia.txt`. Keberadaan file `file_rahasia.txt.asc` menunjukkan bahwa proses enkripsi berhasil dilakukan. Isi file

tersebut sudah tidak dapat dibaca secara langsung karena telah diubah menjadi data terenkripsi. Hanya pemilik kunci privat yang sesuai dengan kunci publik yang digunakan saat proses enkripsi yang dapat membuka dan membaca isi pesan tersebut.

Berdasarkan hasil pengujian, proses enkripsi berhasil dilakukan dan menghasilkan file terenkripsi yang siap dikirim kepada penerima dengan tingkat keamanan yang lebih tinggi dibandingkan file teks biasa.

F.



```
Haza VirtualBox [Berjalan] - Oracle VM VirtualBox
22 Mei 19:31
haza@haza-VirtualBox: ~/Documents
haza@haza-VirtualBox:~/Documents$ gpg --decrypt file_rahasia.txt.asc
gpg: encrypted with 3072-bit RSA key, ID 4A3F21B4, created 2024-06-18
"Budi Santoso <budi@gmail.com>"

Halo Budi,
Ini adalah pesan rahasia untuk tugas GPG.
Semoga berhasil!

haza@haza-VirtualBox:~/Documents$
```

Pada tahap ini dilakukan proses **dekripsi** terhadap file yang telah dienkripsi sebelumnya. Dekripsi merupakan proses mengembalikan data terenkripsi ke bentuk aslinya sehingga isi pesan dapat dibaca kembali oleh penerima yang memiliki kunci privat yang sesuai.

Proses dekripsi dilakukan menggunakan perintah:

```
gpg --decrypt file_rahasia.txt.asc
```

Perintah tersebut digunakan untuk membuka file file_rahasia.txt.asc yang sebelumnya telah dienkripsi menggunakan kunci publik penerima. Saat perintah dijalankan, GPG secara otomatis mencari dan menggunakan kunci privat yang sesuai untuk melakukan dekripsi.

Berdasarkan hasil yang ditampilkan pada terminal, sistem memberikan informasi:

```
gpg: encrypted with 3072-bit RSA key, ID 4A3F21B4, created 2024-06-18
"Budi Santoso <budi@gmail.com>"
```

Informasi tersebut menunjukkan bahwa file telah dienkripsi menggunakan kunci RSA 3072-bit milik **Budi Santoso** dengan alamat email **budi@gmail.com**. Setelah proses dekripsi berhasil dilakukan, isi pesan asli ditampilkan kembali pada terminal, yaitu:

Halo Budi,

Ini adalah pesan rahasia untuk tugas GPG.

Semoga berhasil!

Munculnya isi pesan tersebut membuktikan bahwa file berhasil didekripsi dan dikembalikan ke bentuk semula. Hanya pengguna yang memiliki pasangan kunci privat yang sesuai dengan kunci publik yang digunakan saat proses enkripsi yang dapat membaca isi pesan tersebut.

Berdasarkan hasil pengujian, proses dekripsi berhasil dilakukan tanpa kendala. Pesan yang sebelumnya tersimpan dalam bentuk terenkripsi berhasil dibuka kembali dan dapat dibaca oleh penerima yang berwenang. Hal ini menunjukkan bahwa mekanisme kriptografi asimetris pada GPG bekerja dengan baik dalam menjaga kerahasiaan dan keamanan data selama proses pengiriman pesan.